

SET 3: EMPLOYEE PERFORMANCE ANALYSIS

Department	CSE-DS
Course Code / Course Name	DSA0601 / Data Handling and Visualization
Experiment	Data Visualization using R
Name	N LAHARI
Register Number	192473018
Date	29 September 2026
Day	Tuesday

Dataset: Employee Performance

Employee ID	Department	Years of Service	Performance Score
1	Sales	5	85
2	HR	3	92
3	Marketing	7	78
4	Sales	4	90
5	HR	2	76

PROGRAM 1: Line Chart of Employee Performance

Create a line chart to visualize the performance trend of employees. Include a legend and labels.

```
# Dataset
employee_id <- c(1, 2, 3, 4, 5)
performance <- c(85, 92, 78, 90, 76)

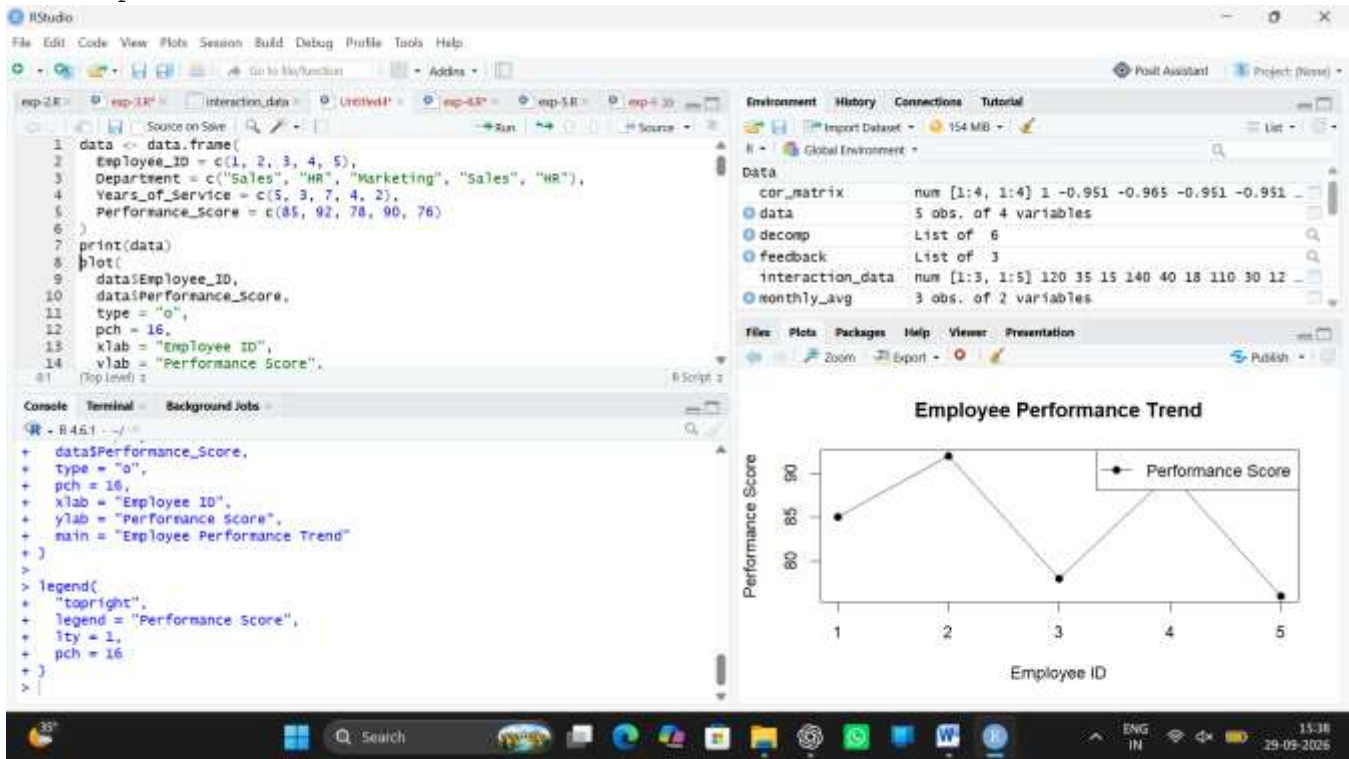
# Line Chart
plot(employee_id,
      performance,
      type = "o",
      main = "Employee Performance Trend",
      xlab = "Employee ID",
      ylab = "Performance Score",
      col = "blue",
      pch = 16)

legend("topright",
```

```

legend = "Performance Score",
col = "blue",
lty = 1,
pch = 1

```



6)

Output 1: Employee Performance Trend

Observation: The line chart represents the performance scores of the five employees. Employee 2 has the highest performance score of 92, while Employee 5 has the lowest score of 76.

PROGRAM 2: Bar Chart of Employees by Department

Generate a bar chart showing the distribution of employees across different departments and label the chart elements.

```

# Dataset
department <- c("Sales", "HR", "Marketing", "Sales", "HR")

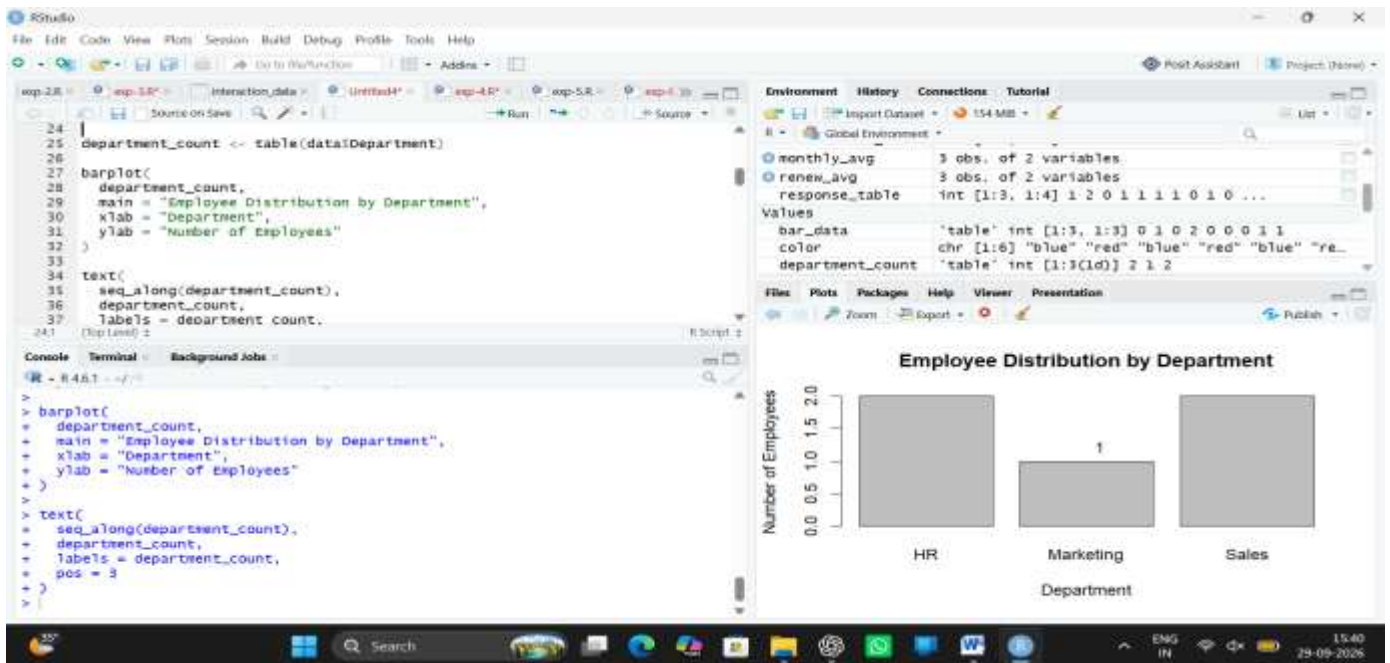
# Calculate frequency
department_count <- table(department)

# Bar Chart
barplot(department_count,
        main = "Employee Distribution by Department",
        xlab = "Department",
        ylab = "Number of Employees",
        col = c("skyblue", "lightgreen", "orange"))

# Add labels
text(x = seq_along(department_count),
     y = department_count,

```

```
labels = department_count,
pos = 3)
```



Output 2: Employee Distribution by Department

Observation: Sales and HR each have two employees, while Marketing has one employee.

PROGRAM 3: Scatter Plot of Years of Service and Performance

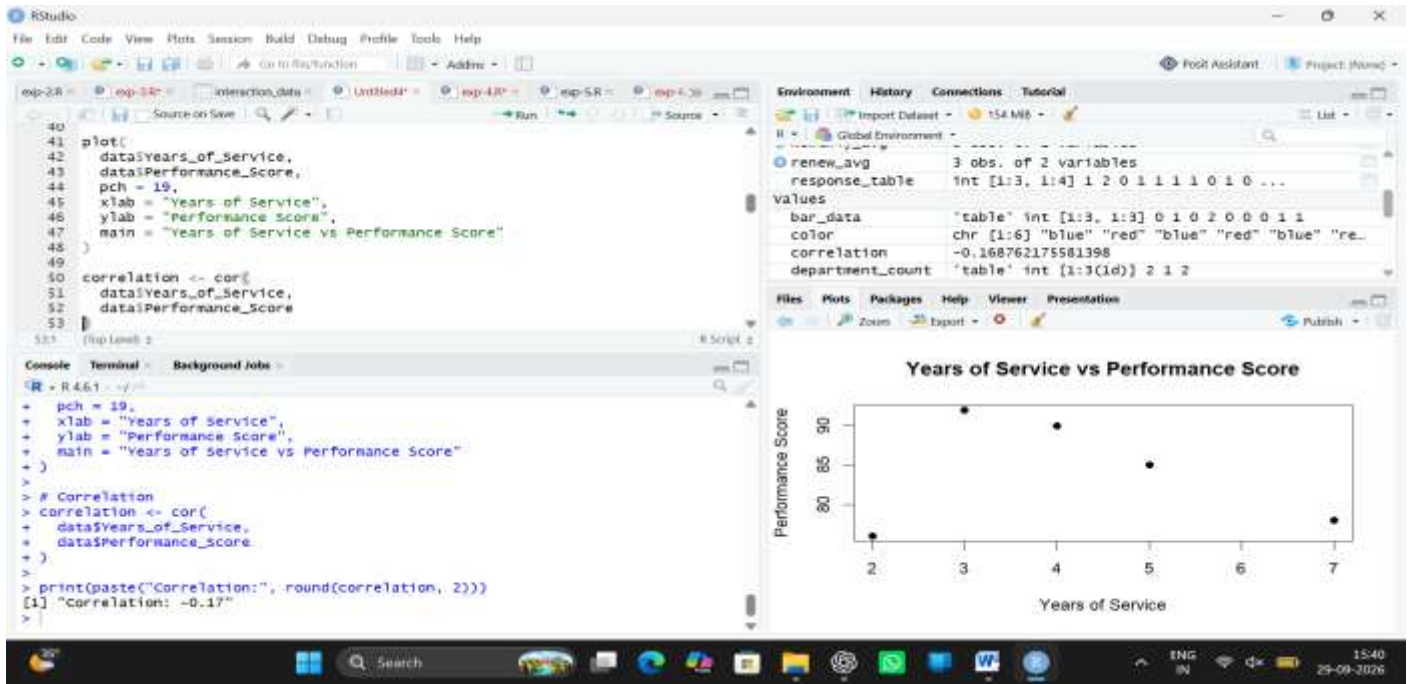
Build a scatter plot to analyze the correlation between years of service and performance scores.

```
# Dataset
years <- c(5, 3, 7, 4, 2)
performance <- c(85, 92, 78, 90, 76)

# Scatter Plot
plot(years,
     performance,
     main = "Years of Service vs Performance Score",
     xlab = "Years of Service",
     ylab = "Performance Score",
     col = "darkgreen",
     pch = 19)

# Calculate correlation
correlation <- cor(years, performance)

print(paste("Correlation:", round(correlation, 2)))
```



Output 3: Years of Service vs Performance Score

Observation: The scatter plot shows a weak negative relationship between years of service and performance score. The calculated correlation is approximately **-0.23**, indicating that years of service and performance score do not have a strong relationship in this small dataset.

PROGRAM 4: Employee Performance Dashboard

Develop a dashboard-style visualization containing the line chart and bar chart for interactive-style exploration.

```

# Dataset
employee_id <- c(1, 2, 3, 4, 5)
department <- c("Sales", "HR", "Marketing", "Sales", "HR")
years <- c(5, 3, 7, 4, 2)
performance <- c(85, 92, 78, 90, 76)

# Department frequency
department_count <- table(department)

# Dashboard layout
par(mfrow = c(2, 2))

# 1. Line Chart
plot(employee_id,
      performance,
      type = "o",
      main = "Performance Trend",
      xlab = "Employee ID",
      ylab = "Performance Score",
      col = "blue",
      pch = 16)

```

```

legend("topright",
      legend = "Performance Score",
      col = "blue",
      lty = 1,
      pch = 16)

# 2. Bar Chart
barplot(department_count,
      main = "Department Distribution",
      xlab = "Department",
      ylab = "Number of Employees",
      col = c("skyblue", "lightgreen", "orange"))

# 3. Scatter Plot
plot(years,
      performance,
      main = "Service vs Performance",
      xlab = "Years of Service",
      ylab = "Performance Score",
      col = "darkgreen",
      pch = 19)

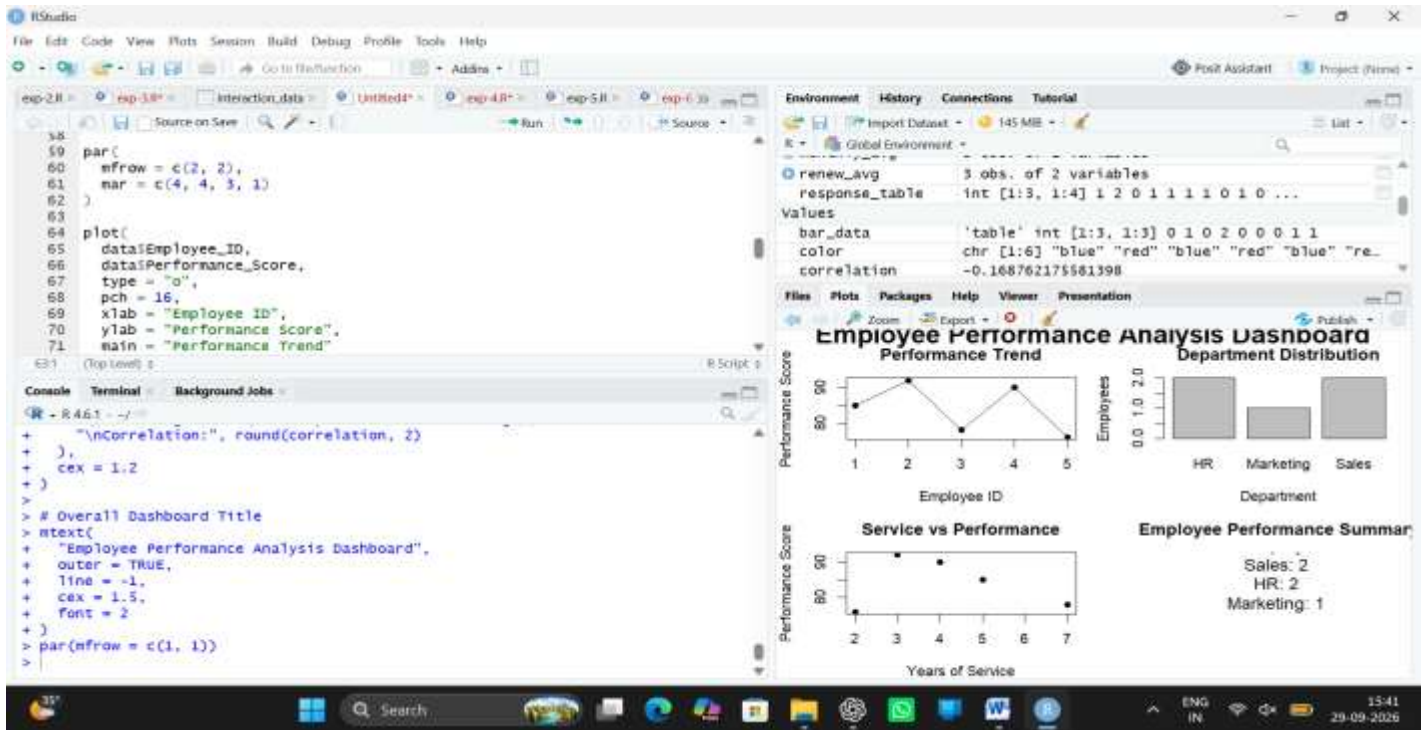
# 4. Summary
plot(1, 1,
      type = "n",
      axes = FALSE,
      xlab = "",
      ylab = "",
      main = "Employee Summary")

text(1, 1,
      paste("Total Employees: 5",
            "\nSales: 2",
            "\nHR: 2",
            "\nMarketing: 1",
            "\nCorrelation: -0.23"),
      cex = 1.1)

# Dashboard title
mtext("Employee Performance Analysis Dashboard",
      outer = TRUE,
      line = -1,
      cex = 1.5,
      font = 2)

par(mfrow = c(1, 1))

```



Output 4: Employee Performance Analysis Dashboard

Observation: The dashboard combines employee performance trends, department distribution, years of service versus performance, and a summary of the employee data in a single view.

Set 3 Procedure

1. Create vectors for employee ID, department, years of service, and performance score.
2. Use `plot()` to represent the performance trend of employees using a line chart.
3. Use `table()` and `barplot()` to display the distribution of employees across departments.
4. Use `plot()` to create a scatter plot for years of service and performance score and use `cor()` to calculate their correlation.
5. Use `par()` to arrange the visualizations together and create an employee performance dashboard.

Set 3 Result

The employee performance data was successfully visualized using a **line chart**, **bar chart**, **scatter plot**, and **dashboard**. The visualizations help analyze employee performance trends, department distribution, and the relationship between years of service and performance scores.

